

User Study – Manual Python Workflow (Method 1 Only)

Instructions: Fill Section A once. For Section B, answer *per use-case*. Duplicate Section B if you completed multiple use-cases.

Participant ID: _____ Your assignment: ☐ Method 1 (Manual Python)

Use case(s) you worked on: ☒ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

Section A – Background (answer once)

A1. Educational level (*highest completed*)

☐ High School ☐ Vocational/Technical ☒ Bachelor's ☐ Master's ☐ Doctorate ☐ Other: _____

A2. Primary role (*pick the closest*)

☐ IoT/CPS Developer ☒ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☐ Software Architect / Full-stack ☐ System Integrator ☐ Other: _____

A3. Self-rated expertise (*tick one per row*)

	Beginner	Intermediate	Advanced	Expert
Python programming	☐	☒	☐	☐
Time-series / ML	☐	☒	☐	☐
Domain knowledge (your use-case)	☐	☒	☐	☐

A4. Prior experience (*used ML frameworks/libraries before?*) ☒ Yes ☐ No

Section B – Task-Specific (duplicate this page *per use-case*)

This B-section is for: ☒ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

B1. Time & Effort

Record your own times. Estimate if needed. “Extra learning time” = time spent reading docs/tutorials.

B1. Time spent (minutes)

	Method 1 (Manual Python)
Writing/testing code (end-to-end)	
Extra learning time	535 min

B2. Lines of code (LoC) (*exclude blanks/comments*)

Manual Python LoC: _____

B2. Task Completion & Ease of Use

Completion: None = could not finish; Partial = finished some parts; Complete = all parts.

B3. Were you able to finish the use-case with Manual Python?

Method 1 (Manual Python): ☐ None ☐ Partial ☒ Complete

B4. Ease by workflow step (Manual Python) (1=Very easy, 5=Very difficult; tick one per cell)

Step	1	2	3	4	5
Whole workflow	☐	☐	☒	☐	☐
Data preprocessing	☒	☐	☐	☐	☐
Feature engineering (lags/rolling)	☐	☒	☐	☐	☐
Model training	☐	☒	☐	☐	☐
Model evaluation/plots	☐	☒	☐	☐	☐

B1. Which steps did you complete? (check all that apply)

- ☒ Preprocessing
- ☒ Model configuration
- ☒ Training
- ☒ Evaluation
- ☒ Plotting/Diagnostics

B2. How difficult was each step? (1=Very easy, 5=Very difficult; tick one per row)

Step	1	2	3	4	5
Preprocessing	☒	☐	☐	☐	☐
Model configuration	☐	☒	☐	☐	☐
Training	☐	☒	☐	☐	☐
Evaluation	☐	☒	☐	☐	☐
Plotting/Diagnostics	☐	☒	☐	☐	☐

B3. Notes (issues, decisions, or gaps to revisit):

B5. Time spent per step (record for this use-case; minutes or start/end)

Step	Start (hh:mm)	End (hh:mm)	Minutes
Preprocessing	_____	_____	180 min
Model configuration	_____	_____	100 min
Training	_____	_____	120 min
Evaluation (metrics/baseline)	_____	_____	45 min
Plotting / Diagnostics	_____	_____	90 min
Debugging (errors, crashes, outputs)	In each step, considered		_____
Extra learning / searching / docs	_____	_____	_____
Total			9 hours

Notes: “Learning” includes reading docs, web searches, tutorials, setup notes, and guides. “Debugging” covers error fixing, crashes, failed runs, and investigating unexpected results.

B6. Comments (bugs, friction points, ideas):

LSTM Train: 245 s Test: 5 s

GRU Train: 238 s Test: 3 s
